

ClaimIQ

AI-Assisted Motor Claims Decision Workbench

Product Requirements Document — MVP v2.0

Capstone: Digital Transformation & Artificial Intelligence

Application Type: Working AI-enabled web application

Primary Cloud Stack: Microsoft Azure AI + Vercel

1. Product Vision

ClaimIQ is an AI-assisted motor insurance claims decision-support application.

It converts an unstructured motor insurance claim into a structured, review-ready decision by:

1. Reading workshop estimates and claim documents.
2. Extracting claim, vehicle, garage and repair information.
3. Interpreting and standardising repair line items using a foundation model.
4. Applying the correct policy rules to calculate an indicative admissible settlement.
5. Identifying repair-estimate inconsistencies and items requiring verification.
6. Routing the claim to the appropriate level of human review.
7. Determining whether specialist or SIU referral is required.
8. Explaining policy- and SOP-dependent decisions using retrieval-augmented generation.
9. Identifying the human approval authority required.
10. Maintaining an auditable record of AI recommendations, rules and human decisions.

ClaimIQ is **not an autonomous claims settlement system**.

The final decision always remains with an authorized human claims professional.

2. Capstone Fit

The capstone requires a working deployed application solving a genuine business pain point with real AI capability, measurable output and an interface suitable for non-technical decision makers. It explicitly allows synthetic data while emphasizing that the AI logic and UX must be real and that narrow, reliable applications are preferred over broad but unstable applications.

ClaimIQ therefore focuses on one end-to-end use case:

Reduce the effort and inconsistency involved in assessing private motor own-damage claims by combining AI document understanding, policy-aware calculations, exception detection and evidence-based claims routing.

3. Primary Business Problem

Claims officers currently spend substantial time on:

- reading workshop estimates,
- manually identifying relevant claim information,
- interpreting abbreviations and inconsistent repair descriptions,
- checking whether repairs correspond with the reported accident,
- locating relevant policy clauses,
- applying depreciation and deductibles,
- checking add-on applicability,
- verifying unusual repair costs,
- determining whether a surveyor or specialist is required,
- escalating suspicious claims,
- explaining deductions,
- documenting decisions.

The result is:

- longer turnaround time,
- inconsistent assessments,
- high manual effort,
- unnecessary surveyor involvement,
- missed leakage signals,
- poor decision traceability.

ClaimIQ redesigns this workflow around **AI-assisted exception management**.

4. Product Proposition

ClaimIQ converts an unstructured motor claim into a review-ready decision by extracting the claim, applying the selected policy, identifying unusual repairs, routing the claim under insurer SOPs and grounding every material decision in source evidence.

5. Primary User

Claims Officer

The primary user should be able to answer:

- What happened?
- What vehicle is involved?
- What is the garage claiming?
- What does the selected policy cover?
- What depreciation or deductible applies?
- What is the indicative admissible settlement?
- What items require verification?
- Does the claim qualify for Fast Track?
- Does a surveyor or specialist need to review it?
- Is SIU referral warranted?
- Who must authorize the claim?
- Which policy/SOP rule drove the decision?

Secondary users:

- Claims Manager
- Motor Surveyor
- Technical Reviewer
- SIU Analyst
- Senior Claims Authority

The MVP does not implement user authentication or role-based access control.

6. Supported Knowledge Corpus

ClaimIQ will use four preloaded synthetic documents.

Policy A

AsterDrive Comprehensive Private Car Policy

Code:

AD-COMP-01

Standard comprehensive product covering:

- own damage,

- depreciation,
- deductible,
- IDV,
- CTL,
- repair vs replacement,
- exclusions,
- settlement mechanics.

The document includes stable machine-readable rule IDs such as `PA-DEP-PLASTIC`, `PA-DEP-AGE`, `PA-PAINT`, `PA-CTL` and `PA-DED`.

Policy B

AsterDrive Elite Protect Private Car Policy

Code:

`AD-ELITE-02`

Includes additional benefits such as:

- Zero Depreciation,
- Consumables Protect,
- Engine & Gearbox Protect,
- Invoice Gap,
- Tyre & Rim Protect,
- Key Replacement,
- Roadside Assistance.

The active Policy Schedule determines which benefits actually apply.

Its machine-readable rule IDs include:

- `PB-ZD-LIMIT`
 - `PB-ZD-TL`
 - `PB-CONS`
 - `PB-ENG`
 - `PB-ENG-ESC`
 - `PB-RTI-AGE`
 - `PB-DED`
 - `PB-CTL`
-

Claims SOP A

Motor Own-Damage Claims Handling SOP

Code:

CLM-MTR-01

Used for:

- routine own-damage claims,
 - Fast Track,
 - Desktop Review,
 - document completeness,
 - standard assessment,
 - human override,
 - approval authority.
-

Claims SOP B

Complex & High-Risk Motor Claims Review & SIU Escalation SOP

Code:

CLM-MTR-02

Used for:

- high-severity claims,
 - structural losses,
 - CTL review,
 - material inconsistencies,
 - high-value claims,
 - provider anomalies,
 - specialist assessment,
 - SIU referral.
-

7. Critical Product Design Principles

7.1 Rules before AI

Policy and SOP rules override AI interpretation.

The claims SOP explicitly establishes this operating principle.

7.2 AI interprets; code calculates

LLMs may:

- understand text,
- normalize repair descriptions,
- classify parts,
- explain findings,
- identify semantic inconsistencies.

LLMs must not perform authoritative settlement arithmetic.

7.3 Review signals are not deductions

Benchmark deviations and anomalies should trigger human review.

They must **not automatically reduce the settlement amount**.

The SOP explicitly states that benchmark variance is a review signal and requires a defensible alternative basis before adjustment.

7.4 Specialist review is not SIU referral

ClaimIQ must separate:

technical/severity escalation

from:

integrity/fraud investigation.

The specialist SOP explicitly states that high severity or a high risk score does not establish fraud.

7.5 RAG explains; it does not decide

RAG retrieves policy/SOP evidence.

The deterministic rule engine calculates settlement and routing.

7.6 Humans remain accountable

No AI output can independently:

- approve a claim,
 - repudiate a claim,
 - classify fraud,
 - approve CTL,
 - make payment.
-

8. Product Scope

P0 — Core MVP

1. Claim Intake
 2. Azure Document Extraction
 3. AI Claim Understanding
 4. Policy Schedule
 5. Policy Rule Registry
 6. Settlement Engine
 7. Intelligent Estimate Review
 8. Review Risk Score
 9. Claims Routing
 10. Specialist vs SIU Decision
 11. Required Approval Authority
 12. Policy/SOP RAG
 13. Missing Information Checklist
 14. Human Decision
 15. Audit Timeline
-

P1 — Include if core MVP is stable

1. Decision Dossier PDF
 2. Consolidated Missing-Information Request Generator
-

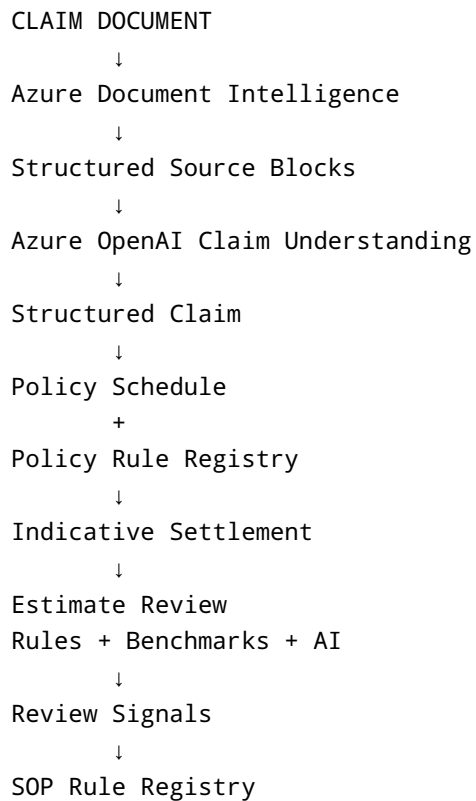
9. Explicitly Out of Scope

Do not build:

- model training,

- fine-tuning,
- claims regression models,
- trained fraud models,
- Isolation Forest,
- SHAP,
- calibrated fraud probabilities,
- dynamic customer policy upload,
- dynamic document indexing during the demo,
- accident-photo analysis,
- image-versus-estimate damage comparison,
- portfolio dashboard,
- garage historical dashboard,
- what-if simulator,
- authentication,
- payment execution,
- core insurer integration,
- third-party bodily injury assessment,
- database infrastructure,
- autonomous agents.

10. Core Application Flow



↓
Fast Track
OR
Desktop Review
OR
Specialist Review
↓
Optional SIU Referral
↓
Required Approval Authority
↓
Human Decision
↓
Audit Trail

Parallel evidence layer:

Policy/SOP documents
↓
Azure AI Search
↓
RAG
↓
Policy/SOP evidence

11. Azure Architecture

The application requires three Azure service families.

12. Azure Document Intelligence

Purpose

Used for:

- OCR,
- text extraction,
- table extraction,
- document structure,
- page layout,
- coordinates,

- source evidence.

Use:

prebuilt-layout

Azure Document Intelligence v4.0 uses the GA 2024-11-30 REST API and supports structured document extraction through SDKs or REST.

Analyze endpoint

```
POST
{DOCUMENT_INTELLIGENCE_ENDPOINT}/documentintelligence/
documentModels/prebuilt-layout:analyze
?api-version=2024-11-30
```

Authentication:

```
Ocp-Apim-Subscription-Key:
{DOCUMENT_INTELLIGENCE_KEY}
```

The response is asynchronous.

Store the returned:

Operation-Location

and poll until analysis succeeds.

13. Azure OpenAI / Microsoft Foundry

Purpose

Used for:

- repair normalization,
- material classification,
- repair/action classification,
- accident interpretation,
- semantic impact review,
- grounded explanations,
- missing-information request generation.

Use the current Azure OpenAI **v1 API**.

The current Microsoft guidance supports the OpenAI SDK against an Azure base URL ending in `/openai/v1/`, and the Responses API is recommended for new applications.

Base URL:

```
https://{AZURE_RESOURCE}.openai.azure.com/openai/v1/
```

Responses endpoint:

```
POST /responses
```

Do not hard-code a specific foundation-model version.

Configure:

```
AZURE_OPENAI_CHAT_DEPLOYMENT
```

The selected deployment must:

- support the Responses API,
- support Structured Outputs,
- be generally available in the selected Azure region.

Azure currently supports schema-constrained Structured Outputs through the Responses API using `text.format`.

14. Azure OpenAI Embeddings

Deploy:

```
text-embedding-3-small
```

or an equivalent supported embedding model.

Microsoft currently supports `text-embedding-3-small` and documents the v1 embeddings API for generating vectors.

Endpoint:

```
POST /embeddings
```

Use the same deployment for:

- knowledge-document vectors,
- runtime question vectors.

15. Azure AI Search

Purpose

Used for RAG retrieval across:

- Policy A,
- Policy B,
- SOP A,
- SOP B.

Azure AI Search supports combined full-text and vector retrieval in a single hybrid request; results are merged using Reciprocal Rank Fusion.

Use API:

```
2026-04-01
```

which Microsoft currently identifies as the latest stable Search REST API.

For the prototype, the Free tier is suitable for a small development corpus if available in the subscription.

16. Azure Resources to Create

Resource A

Azure Document Intelligence

Suggested name:

```
claimiq-document-intelligence
```

Environment:

```
AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT  
AZURE_DOCUMENT_INTELLIGENCE_KEY
```

Resource B

Azure OpenAI / Microsoft Foundry

Suggested resource:

```
claimiq-foundry
```

Deploy:

Chat

```
claimiq-chat
```

Embedding

```
claimiq-embedding
```

Environment:

```
AZURE_OPENAI_ENDPOINT  
AZURE_OPENAI_API_KEY  
AZURE_OPENAI_CHAT_DEPLOYMENT  
AZURE_OPENAI_EMBEDDING_DEPLOYMENT
```

Resource C

Azure AI Search

Suggested name:

```
claimiq-search
```

Index:

```
claimiq-knowledge
```

Environment:

```
AZURE_SEARCH_ENDPOINT  
AZURE_SEARCH_ADMIN_KEY  
AZURE_SEARCH_QUERY_KEY  
AZURE_SEARCH_INDEX
```

17. Policy Schedule

Policy wording alone is insufficient.

Every claim must contain an active **Policy Schedule**.

Example:

```
type PolicySchedule = {  
  policyNumber: string;  
  
  policyCode:  
    | "AD-COMP-01"  
    | "AD-ELITE-02";  
  
  policyStartDate: string;  
  policyEndDate: string;  
  
  vehicleRegistration: string;  
  engineCC: number;  
  
  idv: number;  
  
  compulsoryDeductible: number;  
  voluntaryDeductible: number;  
  
  activeAddOns: AddOn[];  
  
  odClaimsThisPolicyYear: number;  
  
  originalInvoiceValue?: number;  
  
  benefitUsage?: {  
    engineProtectUsed?: number;  
    tyreProtectUsed?: number;  
    keyReplacementUsed?: number;
```

```
    rsaEventsUsed?: number;  
  };  
};
```

Policy A explicitly establishes that IDV, deductible, vehicle identity and special conditions belong in the Policy Schedule.

Policy B states that an add-on should not be assumed active merely because it exists in the product wording.

18. Policy Selector UI

At claim creation:

Active Policy

AsterDrive Comprehensive

or

AsterDrive Elite Protect

Display:

- Policy number
- Policy type
- IDV
- Deductible
- Active add-ons
- Claims used under add-on limits

Example:

AsterDrive Elite Protect

IDV ₹12.50L

Zero Dep: Active — 1/2 claims used

Consumables: Active

Engine Protect: Active

19. Rule Registry

Create:

```
/lib/rules/ruleRegistry.ts
```

Every deterministic application rule must have:

```
type RuleDefinition = {  
  ruleId: string;  
  
  documentCode: string;  
  
  documentType:  
    | "POLICY"  
    | "SOP";  
  
  clauseNumber: string;  
  
  category: string;  
  
  description: string;  
  
  logic: unknown;  
};
```

Example:

```
{  
  ruleId: "PA-DEP-PLASTIC",  
  documentCode: "AD-COMP-01",  
  documentType: "POLICY",  
  clauseNumber: "4.1",  
  category: "DEPRECIATION",  
  description:  
    "50% depreciation on plastic/rubber/nylon parts",  
  logic: {  
    type: "percentage",  
    value: 0.50  
  }  
}
```

20. Traceability Principle

Every decision should have:

Decision



Rule ID



Document



Clause

Example:

Plastic depreciation

-₹9,000

Rule:

PA-DEP-PLASTIC

Policy:

AD-COMP-01

Clause:

4.1

21. Module 1 — Intelligent Claim Intake

Input

Accept:

- PDF
- PNG
- JPG
- JPEG

Primary document:

Workshop estimate / job card

Optional:

Claim form

22. Document Viewer

Display the original estimate.

Features:

- zoom,
- page navigation,
- fit width,
- source highlighting.

Right-hand panel:

Extracted Claim

Repair Table

23. Source Blocks

Convert Document Intelligence output into:

```
type SourceBlock = {  
  blockId: string;  
  
  pageNumber: number;  
  
  text: string;  
  
  type:  
    | "paragraph"  
    | "line"  
    | "table_cell";  
  
  polygon?: number[];  
  
  rowIndex?: number;  
  
  columnIndex?: number;  
};
```

Do not immediately flatten the Azure response.

Preserve document coordinates.

24. Module 2 — AI Claim Understanding

Document Intelligence determines:

What exists in the document?

Azure OpenAI determines:

What does it mean?

The model receives compact source blocks.

It returns structured claim data.

25. Structured Claim

```
type StructuredClaim = {  
  claim: {  
    claimNumber?: string;  
    policyNumber?: string;  
    lossDate?: string;  
  
    accidentDescription?: string;  
  
    primaryImpactLocation?: string;  
  };  
  
  vehicle: {  
    registration?: string;  
    make?: string;  
    model?: string;  
    variant?: string;  
  
    manufacturingYear?: number;  
  
    idv?: number;  
  };  
  
  garage: {  
    name?: string;  
    city?: string;  
  };  
};
```

```
    networkStatus:
      | "network"
      | "non_network"
      | "unknown";
  };

  estimate: {
    estimateNumber?: string;
    partsTotal?: number;
    labourTotal?: number;
    paintTotal?: number;
    grandTotal?: number;
  };

  lineItems: RepairLineItem[];

  missingInformation: MissingItem[];
};
```

26. Repair Line Item

```
type RepairLineItem = {
  id: string;

  rawDescription: string;

  normalizedDescription: string;

  quantity?: number;

  partCost?: number;
  labourCost?: number;
  paintCost?: number;
  totalCost?: number;

  category:
    | "part"
    | "labour"
    | "paint"
    | "other";

  material:
    | "metal"
    | "plastic"
```

```

    | "glass"
    | "fiberglass"
    | "rubber"
    | "electrical"
    | "mechanical"
    | "paint"
    | "unknown";

  action:
    | "repair"
    | "replace"
    | "paint"
    | "inspect"
    | "unknown";

  vehicleLocation?: string;

  sourceBlockIds: string[];

  confidence:
    | "high"
    | "medium"
    | "low";
};

```

27. AI Source Mapping

The foundation model must never invent document coordinates.

It returns:

```
sourceBlockIds
```

The frontend obtains polygons from the original Azure output.

Example:

```

{
  "normalizedDescription":
    "Front bumper assembly",

  "sourceBlockIds": [

```

```
"P1-T2-R4-C1",  
"P1-T2-R4-C5"  
]  
}
```

28. Bi-Directional Highlighting

Repair Table → Document

Hover/click repair row.

Highlight corresponding source blocks.

Document → Repair Table

Click source region.

Highlight and scroll to corresponding repair row.

This is a core demo feature.

29. Human Correction

Extracted values remain editable.

When manually corrected:

Display:

Human Verified

Do not imply corrections retrain the model.

30. Document Completeness

Calculate a deterministic completeness score.

Example required fields:

- active policy,
- vehicle registration,
- accident description,
- workshop estimate,
- repair items,
- total estimate.

Optional/conditional:

- licence,
- police record,
- photos,
- diagnostics,
- invoice.

Display:

Claim Completeness

82%

31. Missing Information Panel

Example:

Needs Attention

- Accident description incomplete
- Damage photographs unavailable
- Driving licence not provided

The standard claims SOP explicitly instructs that missing information should preferably be requested together rather than through repeated ad hoc requests.

Optional button:

Generate Consolidated Request

Azure OpenAI generates a concise request using only the identified missing items.

32. Module 3 — Policy-Aware Settlement Engine

Settlement is deterministic.

Azure OpenAI must not calculate the final settlement.

33. Settlement Sequence

```
Quoted Repair Cost
    ↓
Covered Accident-Related Cost
    ↓
Remove Non-Covered Items
    ↓
Apply Depreciation
    ↓
Apply Eligible Add-Ons
    ↓
Apply Deductible
    ↓
Apply Applicable Limits
    ↓
Indicative Admissible Settlement
```

34. Policy A — Depreciation Rules

Policy A establishes:

- plastic/rubber/nylon: 50%
- fiberglass: 30%
- glass: nil
- other parts: age-based depreciation,
- painting: depreciation applies to paint material.

Implement directly from rule IDs.

35. Policy A — Age Depreciation

≤6 months	0%
>6m-1 year	5%
>1-2 years	10%
>2-3 years	15%
>3-4 years	25%
>4-5 years	35%
>5-10 years	40%
>10 years	50%

36. Policy A — Painting

When paint material is separately available:

50% depreciation
on material component

When paint charge is consolidated:

Material component =
25% of total painting charge

then depreciation applies to the derived material amount.

37. Policy B — Zero Depreciation

Policy B provides Zero Depreciation for the first two admissible partial-loss claims, subject to active coverage. It does not waive deductibles/non-covered items and does not apply to TL/CTL.

The engine must therefore evaluate:

zeroDepActive
AND
claimType = partial_loss

AND
OD claims used < 2

before waiving relevant depreciation.

38. Policy B — Consumables

Where active:

eligible repair consumables may be admitted up to:

₹15,000 per claim

subject to policy conditions.

39. Policy B — Engine Protect

Prototype cap:

₹1,50,000 per policy year

Specialist trigger:

- estimated benefit >₹75,000,
 - or complete engine/gearbox replacement.
-

40. Settlement Output

Display:

Workshop Estimate

₹1,26,500

Indicative Admissible Settlement

₹98,700

Policy Adjustment

₹27,800

41. Settlement Bridge

Example:

Driver	Impact
Workshop estimate	₹1,26,500
Non-covered item	-₹6,500
Plastic depreciation	-₹8,500
Age depreciation	-₹9,800
Deductible	-₹2,000
Indicative settlement	₹99,700

42. Policy Adjustment vs Amount Under Review

Keep separate.

Policy Adjustments

Automatically change settlement because the rule is encoded.

Examples:

- depreciation,
- deductible,
- add-on cap,
- exclusion.

Amount Under Review

Does **not** automatically change settlement.

Examples:

- suspicious rear lamp,

- high labour cost,
- possible duplicate part,
- repairability question.

UI:

Indicative Settlement

₹99,700

Amount Requiring Verification

₹22,400

43. CTL Logic

Policy A and B use a contractual CTL trigger above 75% of IDV.

The specialist SOP adds an internal warning stage at 60% of IDV.

Therefore:

Repair / IDV <60%

Normal partial-loss assessment.

≥60%

CTL Watch

Route to Specialist.

>75%

CTL Candidate

Stop normal partial-loss workflow.

Require authorized CTL assessment.

44. CTL UI

Show:

Repair Cost / IDV

64%

Status:

⚠ CTL Watch

or:

79%

Status:

● CTL Candidate

45. Module 4 — Intelligent Estimate Review

Purpose:

Identify elements requiring human verification.

Three layers.

46. Layer A — Deterministic Checks

Implement:

- duplicate line,
 - repeated part number,
 - duplicate labour operation,
 - arithmetic inconsistency,
 - missing amount,
 - high estimate/IDV,
 - high replacement intensity,
 - identity mismatch,
 - impossible chronology.
-

47. Arithmetic Anomaly

Trigger where estimate reconciliation differs by:

> ₹5,000
OR
> 2%

using the applicable SOP rule.

48. Duplicate Signal

If duplicate or near-duplicate repair lines change gross estimate by:

>₹10,000

route to specialist review.

49. Layer B — Synthetic Benchmarking

Create:

```
/lib/benchmarks/repairBenchmarks.ts
```

Dimensions:

- vehicle class,
- city tier,
- network status.

Metrics:

- labour rate,
- paint cost/panel,
- replacement intensity,
- parts/labour ratio,
- estimate/IDV.

All benchmark panels must display:

50. Provider Review Signals

Use SOP thresholds.

Examples:

Labour Rate

40% above applicable benchmark

→ verification / specialist review if material.

Paint

50% above benchmark across 2+ panels

→ request breakdown.

Replacement Intensity

70% replacement lines

→ evaluate repairability.

Potential Excess

₹25,000

→ specialist review.

51. Layer C — AI Semantic Review

Input:

- accident narrative,
- primary impact location,
- repair lines.

Azure OpenAI returns:

CONSISTENT

POSSIBLY_CONSISTENT

REQUIRES_VERIFICATION

INCONSISTENT_WITH_DESCRIPTION

52. Semantic Review Rules

Example:

Accident:

Front-right impact

Repair:

Rear-left tail lamp

AI:

Requires verification because the repair location does not directly correspond with the stated primary impact.

The SOP requires mismatches to be verified before adverse action.

53. Semantic Mismatch Routing

Single mismatch $\leq$ ₹10,000

Desktop clarification.

Mismatch $>$ ₹10,000

Specialist review.

Safety-critical mismatch

Specialist review.

Multiple geographic mismatches

Specialist review.

Mismatch + separate integrity concern

Evaluate SIU rules.

54. Review Risk Score

Create a transparent score from 0–100.

Suggested weights:

Driver	Weight
Estimate benchmark deviation	25
Replacement intensity	15
Semantic inconsistency	25
Duplicate/arithmetic signals	15
Data/identity quality	10
Claim severity	10

This score is an internal deterministic measure.

It is **not a fraud probability**.

55. Risk Bands

0–29	Low
30–59	Moderate
60–79	High
80–100	Very High

56. Module 5 — Claims Routing

Primary route has exactly three states.

Green

Fast Track Candidate

Amber

Desktop Surveyor Review

Red

Specialist Review

SIU is a secondary escalation decision.

57. Routing Precedence

Specialist Hard Trigger?

↓ YES

Specialist Review

↓ NO

Desktop Hard Trigger?

↓ YES

Desktop Review

↓ NO

Fast Track Conditions Met?

↓ YES

Fast Track

↓ NO

Desktop Review

58. Fast Track Requirements

All must be satisfied:

Indicative settlement $\leq$ ₹60,000

Review Risk $<$ 30

Document completeness $\geq$ 90%

No Desktop trigger

No Specialist trigger

These thresholds are defined in SOP A.

59. Desktop Review Triggers

Include:

- risk 30–59,
 - estimate/IDV $>$ 15%,
 - non-network labour $>$ 30% above benchmark,
 - low-confidence field changes settlement $>$ ₹10,000,
 - safety-critical repair ambiguity,
 - policy interpretation required.
-

60. Specialist Triggers

Include:

- Review Risk $\geq$ 60,
- settlement $>$ ₹2.5L,
- repair/retrieval $\geq$ 60% IDV,
- rollover,
- major structural damage,
- multiple airbag deployment,
- fire with material damage,
- flood + engine/gearbox internal damage,
- major powertrain replacement,
- significant impact mismatch,

- material duplicate charges,
 - serious identity/document inconsistency.
-

61. Specialist vs SIU

Specialist Review answers:

Does this claim require higher technical or claims expertise?

SIU answers:

Does this claim contain sufficient unresolved integrity concerns to justify investigation?

Do not combine them.

62. SIU Decision

Display separately:

SIU Referral

Not Required

or

Recommended

63. SIU Referral Rules

SIU should ordinarily require:

Either

one confirmed critical integrity trigger,

Or

at least two independent unresolved material concern categories.

64. Critical Integrity Examples

Examples include:

- confirmed forged/tampered material document,
 - confirmed vehicle-identity manipulation,
 - credible evidence of staged/collusive loss,
 - material false statement remaining after clarification.
-

65. AI Risk Alone Cannot Trigger SIU

A high ClaimIQ Review Risk score alone must never trigger:

- SIU,
- fraud conclusion,
- repudiation.

This is explicitly stated in the specialist SOP.

66. Routing Output

Example:

Specialist Review

Review Risk: 72 / 100

Key Reasons

- settlement above ₹2.5L,
- labour 48% above benchmark,
- rear-left repair inconsistent with documented front impact.

SIU Referral

Not currently indicated

Reason:

Technical and estimate concerns exist, but the current evidence does not satisfy SIU referral criteria.

67. Required Approval Authority

ClaimIQ determines the required authorization level.

SOP A establishes prototype approval bands:

≤ ₹1,50,000

Claims Officer

₹1,50,001–₹5,00,000

Claims Manager

> ₹5,00,000

Senior Claims Authority

68. Specialist Governance

Examples:

- CTL classification → Claims Manager + authorized technical/survey input.
 - SIU referral → Claims Manager.
 - Fraud-related adverse decision → Senior Claims Authority after investigation.
 - AI adverse recommendation → cannot be actioned without human validation.
-

69. Decision Center UI

CLAIM DECISION

Indicative Settlement

₹3,42,800

Amount Under Review

₹38,500

Review Risk

72 / 100

Primary Route

Specialist Review

SIU Referral
Not Required

Approval Required
Claims Manager

70. Module 6 — Policy & SOP RAG

RAG corpus:

AD-COMP-01
AD-ELITE-02
CLM-MTR-01
CLM-MTR-02

Use pre-indexed PDFs.

Do not index during the live demo.

71. Knowledge Ingestion

One-time pipeline:

PDF
↓
Azure Document Intelligence
↓
Section-aware text
↓
Chunking
↓
Azure OpenAI Embeddings
↓
Azure AI Search

72. Chunking

Prefer:

- section,
- clause,
- paragraph,

rather than arbitrary character splitting.

Target:

approximately 400–800 tokens.

Metadata must preserve:

- document,
- clause,
- page,
- rule IDs.

73. Search Index Schema

```
type KnowledgeChunk = {  
  id: string;  
  
  documentCode: string;  
  
  documentVersion: string;  
  
  documentType:  
    | "motor_policy"  
    | "claims_sop";  
  
  policyCode?: string;  
  
  sectionTitle: string;  
  
  clauseNumber?: string;  
  
  pageNumber?: number;  
  
  ruleIds: string[];
```

```
content: string;  
  
contentVector: number[];  
};
```

74. Two RAG Retrieval Modes

Mode A — User Question

Example:

Why has depreciation been applied?

Use:

Hybrid keyword + vector retrieval

Mode B — Application Explanation

Example:

Explain `PA-DEP-PLASTIC`.

Retrieve by:

```
ruleIds contains PA-DEP-PLASTIC
```

This exact-rule lookup should be preferred when ClaimIQ is explaining its own calculation.

75. RAG Question Scope

Policy questions

Filter:

```
documentType = motor_policy
AND
policyCode = activePolicy
```

Routing questions

Filter:

```
documentType = claims_sop
```

76. RAG Runtime

```
User question
  ↓
Embedding
  ↓
Azure AI Search
keyword + vector
  ↓
Top relevant clauses
  ↓
Azure OpenAI
  ↓
Grounded answer
```

77. RAG Answer Structure

Answer

Short explanation.

Calculation

Where relevant.

Source

Policy/SOP name.

Clause

Clause number.

Page

Page number.

Rule

Rule ID where applicable.

78. Example

Why did ClaimIQ deduct ₹9,000?

The bumper has been classified as a plastic replacement component. Policy A applies 50% depreciation to plastic replacement parts.

Calculation

Quoted bumper cost	₹18,000
Depreciation	50%
Deduction	₹9,000

Evidence

Policy:

AsterDrive Comprehensive

Clause:

4.1

Rule:

PA-DEP-PLASTIC

79. RAG Hallucination Guardrail

If no relevant evidence is retrieved:

Display:

No sufficiently relevant policy or SOP evidence was found. Human review is required.

The model must not use generic insurance knowledge to fill the gap.

80. Ask ClaimIQ Drawer

Suggested questions:

- Why was depreciation applied?
 - Why is glass treated differently?
 - Why is this item under review?
 - Why did this claim not qualify for Fast Track?
 - Why was Specialist Review triggered?
 - Does this claim satisfy SIU conditions?
 - Which policy rule drove this deduction?
 - What information is missing?
 - Who must approve the claim?
-

81. Module 7 — Human Decision

ClaimIQ shows:

AI / System Recommendation

Then human actions:

Accept Recommendation

Change Route

Request Additional Evidence

Escalate to Specialist

Refer to SIU

where permitted.

82. Human Override

Require:

Reason Code

Examples aligned with SOP A:

- H01 — Additional physical inspection
- H02 — Policy interpretation
- H03 — Repair methodology
- H04 — Customer evidence
- H05 — Escalation concern
- H06 — Data correction

Free-text comment optional.

83. Audit Timeline

Example:

```
19:04 Claim opened
19:04 Estimate uploaded
19:04 Document Intelligence completed
19:05 16 repair lines identified
19:05 Policy AD-COMP-01 loaded
```

```
19:05 Indicative settlement ₹98,700

19:05 ₹17,500 marked under review

19:05 Review Risk 54/100

19:06 Desktop Review recommended

19:06 Claims Officer accepted routing

19:07 Decision recorded
```

84. Audit Object

```
type AuditEvent = {
  id: string;

  timestamp: string;

  actor:
    | "SYSTEM"
    | "AZURE_DI"
    | "AZURE_OPENAI"
    | "RULE_ENGINE"
    | "USER";

  action: string;

  relatedRuleIds?: string[];

  metadata?: Record<string, unknown>;
};
```

85. Three Demo Claims

Build exactly three.

86. Demo 1 — Hyundai Creta

Purpose

Demonstrate Fast Track.

Characteristics:

- minor parking scrape,
- network garage,
- complete estimate,
- low settlement,
- high document completeness,
- no major review signal.

Expected:

Review Risk <30

Settlement <₹60k

Fast Track Candidate

87. Demo 2 — Honda City

Purpose

Demonstrate Desktop Review.

Characteristics:

- front-right collision,
- bumper,
- headlamp,
- fender,
- paint,
- moderate settlement,
- some repair/replace ambiguity.

Expected:

Moderate Review Risk

Desktop Surveyor Review

88. Demo 3 — BMW 3-Series

Purpose

Demonstrate Specialist decision.

Characteristics:

- front collision,
- premium non-network repairer,
- high labour,
- high replacement intensity,
- duplicate labour,
- rear-left lamp mismatch,
- material amount under review.

Expected:

High Review Risk

Specialist Review

SIU should initially be:

Not automatically required

unless a second independent integrity concern is deliberately introduced.

This is important for demonstrating the difference between **risk** and **fraud investigation**.

89. Application Screens

Keep the MVP to three primary screens.

Screen 1

Claim Intake

Screen 2

Claim Workspace

Screen 3

Decision Dossier

Most interaction occurs on Screen 2.

90. Claim Workspace Layout

CLAIM HEADER

Vehicle | Policy | IDV | Claim Status

DOCUMENT
VIEWER

EXTRACTED ESTIMATE
TABLE

POLICY
SETTLEMENT

CLAIM
ROUTING

AMOUNT
UNDER REVIEW

REVIEW
FINDINGS

POLICY / SOP
EVIDENCE

REQUIRED
AUTHORITY

HUMAN DECISION

AUDIT TIMELINE

91. Claim Header

Show:

- claim ID,
- vehicle,
- registration,
- policy,
- IDV,
- garage,
- network status.

92. Decision Summary Rail

Persistent right-side panel:

DECISION SUMMARY

Settlement

₹98,700

Under Review

₹17,500

Review Risk

54 / 100

Route

Desktop Review

SIU

No

Authority

Claims Officer

93. AI Transparency

Every major output should show its source type.

Use small badges:

AI

Azure OpenAI interpretation.

RULE

Deterministic rule engine.

RAG

Retrieved policy/SOP evidence.

SYNTHETIC

Prototype benchmark.

HUMAN

Human decision.

This makes the architecture visible during the demonstration.

94. API Routes

Recommended Next.js routes:

```
/api/claim/analyze-document
```

```
/api/claim/understand
```

```
/api/claim/semantic-review
```

```
/api/claim/settlement
```

```
/api/claim/review
```

```
/api/claim/routing
```

```
/api/rag/query  
  
/api/human/decision  
  
/api/dossier
```

95. Rules Folder

```
/lib/rules/  
  
ruleRegistry.ts  
  
policyA.ts  
  
policyB.ts  
  
claimsSOPA.ts  
  
claimsSOPB.ts  
  
settlementEngine.ts  
  
reviewRiskEngine.ts  
  
routingEngine.ts  
  
approvalAuthority.ts
```

96. AI Folder

```
/lib/ai/  
  
azureOpenAI.ts  
  
schemas.ts  
  
prompts/  
  
claimUnderstanding.ts
```

```
semanticReview.ts  
ragAnswer.ts  
missingInfoRequest.ts
```

97. RAG Folder

```
/lib/rag/  
azureSearch.ts  
embedding.ts  
retrieval.ts  
chunking.ts  
knowledgeSeeder.ts
```

98. Knowledge Assets

```
/knowledge/  
AD-COMP-01.pdf  
AD-ELITE-02.pdf  
CLM-MTR-01.pdf  
CLM-MTR-02.pdf
```

DOCX versions may be kept separately as editable sources.

PDF should be indexed because stable pages improve citation traceability.

99. Environment Variables

```
# Document Intelligence

AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT=
AZURE_DOCUMENT_INTELLIGENCE_KEY=

# Azure OpenAI

AZURE_OPENAI_ENDPOINT=
AZURE_OPENAI_API_KEY=

AZURE_OPENAI_CHAT_DEPLOYMENT=
AZURE_OPENAI_EMBEDDING_DEPLOYMENT=

# Azure Search

AZURE_SEARCH_ENDPOINT=
AZURE_SEARCH_ADMIN_KEY=
AZURE_SEARCH_QUERY_KEY=
AZURE_SEARCH_INDEX=claimiq-knowledge

# App

DEMO_MODE=false
```

All credentials must remain server-side.

The capstone instructions also explicitly require API keys to be stored securely rather than hard-coded in public-facing application code.

100. Structured Outputs

Use JSON Schema / Structured Outputs for:

- claim understanding,
- semantic review,
- RAG response,
- missing-information request.

Validate again locally with Zod.

Never directly trust uncontrolled model text as application data.

101. Claim Understanding Prompt

Core instruction:

You support motor insurance document interpretation.

Use only the supplied Azure Document Intelligence source blocks.

Never invent missing information.

Normalize repair terminology conservatively.

Map every extracted value to sourceBlockIds.

Classify uncertain material as "unknown".

Do not calculate settlement.

Do not approve or reject a claim.

Return the required structured schema only.

102. Semantic Review Prompt

Compare the supplied accident description, primary impact location and repair estimate.

Classify each relevant repair item as:

CONSISTENT

POSSIBLY_CONSISTENT

REQUIRES_VERIFICATION

INCONSISTENT_WITH_DESCRIPTION

Do not infer fraud.

A mismatch is a review signal only.

Provide a short factual reason.

Return structured output.

103. RAG Prompt

Use only the supplied policy/SOP excerpts.

Do not use general insurance knowledge.

Do not invent rules.

Clearly distinguish:

POLICY RULE

SOP RULE

CLAIMIQ CALCULATION

AI INTERPRETATION

If the evidence is insufficient,
say human review is required.

Always provide the source document,
clause and page where available.

104. Performance Principle

Use AI only where needed.

Do not invoke an LLM for:

- arithmetic,
- depreciation,
- deductibles,
- CTL ratio,
- risk arithmetic,
- routing thresholds,
- approval authority,
- filtering,
- sorting.

These are normal code.

105. Error Handling

Document Intelligence fails

Show:

Document extraction is unavailable. Retry or use a demo claim.

OpenAI fails

Show:

AI interpretation could not complete. Extracted document data remains available.

Search fails

Show:

Policy evidence retrieval is unavailable. Encoded settlement rules remain available.

Structured output invalid

Retry once.

Then:

Human review required.

No RAG evidence

Do not answer from model memory.

106. Demo Resilience

Each demo claim should contain:

```
claim.json
```

```
document.pdf
```

```
cached-di-response.json
```

```
cached-ai-response.json
```

When real Azure works:

● LIVE AI

When cached:

DEMO MODE

Never silently disguise cached responses as live Azure inference.

107. Governance Panel

Show:

AI Role

Decision support.

Settlement

Deterministic rules.

Policy Evidence

RAG.

Risk Signals

Rules + synthetic benchmarks + AI semantic review.

Final Authority

Human claims personnel.

Training Data

No custom model training.

Benchmark Data

Synthetic prototype data.

108. Business KPIs

Position ClaimIQ around:

Claims turnaround time

Less document reading and manual interpretation.

Straight-through processing

Low-risk claims identified faster.

Surveyor productivity

Review concentrated on exceptions.

Claims leakage

Unusual estimate elements surfaced earlier.

Decision consistency

Policy/SOP rules applied systematically.

Auditability

Every major decision linked to evidence.

Do not claim measured percentage improvement without actual production data.

109. Non-Functional Requirements

Reliability

The full BMW demo must work without manual code changes.

Explainability

Every material settlement deduction must display a rule.

Traceability

Every AI extraction should link to document evidence.

Responsiveness

Primary desktop:

1366×768+

Security

No API key in browser.

Privacy

Use synthetic claims only for demonstration.

Accessibility

Risk state must use text plus color.

110. Acceptance Criteria — Document AI

A workshop PDF must:

- upload,
 - call Document Intelligence,
 - extract text/tables,
 - retain coordinates,
 - create repair rows,
 - support row ↔ document highlighting.
-

111. Acceptance Criteria — AI Understanding

The application must:

- normalize line descriptions,
 - classify materials where reasonable,
 - classify repair/replace,
 - identify location,
 - return source block IDs,
 - flag uncertainty.
-

112. Acceptance Criteria — Policy

The selected Policy Schedule must materially change the settlement.

Example:

Same repair estimate:

Policy A

standard depreciation.

Policy B

Zero Dep applies if eligible.

The final amount must differ accordingly.

113. Acceptance Criteria — Settlement

Settlement:

- must be deterministic,
 - must reconcile mathematically,
 - must show every major adjustment,
 - must link each policy adjustment to rule ID,
 - must distinguish under-review amounts.
-

114. Acceptance Criteria — Review

BMW claim must identify:

- elevated labour,
- duplicate operation,
- impact mismatch.

These findings must not automatically reduce settlement.

115. Acceptance Criteria — Routing

Creta:

Fast Track

Honda:

Desktop Review

BMW:

Specialist Review

116. Acceptance Criteria — SIU

A high-risk BMW claim with only technical/estimate concerns:

SIU Not Required

BMW with a second independent unresolved integrity concern:

SIU Referral Recommended

This demonstrates correct governance.

117. Acceptance Criteria — RAG

Question:

Why was depreciation applied?

Must return:

- explanation,
- policy,
- clause,
- page,
- applicable rule ID.

Question:

Why was this escalated?

Must retrieve SOP evidence rather than policy wording.

118. Acceptance Criteria — Human Decision

User can:

- accept recommendation,
 - change route,
 - choose override reason,
 - record comment,
 - create audit event.
-

119. Optional Decision Dossier

PDF should include:

- claim details,
- policy,
- estimate,
- indicative settlement,
- amount under review,
- review findings,
- routing,
- SIU result,
- authority required,
- rule IDs,
- policy/SOP evidence,
- human decision,
- audit summary.

120. Build Sequence

Phase 1

Frontend shell + demo claims.

Phase 2

Azure Document Intelligence.

Phase 3

Azure OpenAI claim understanding.

Phase 4

Policy Schedule + Rule Registry.

Phase 5

Policy A settlement.

Phase 6

Policy B + add-ons.

Phase 7

Estimate review.

Phase 8

Routing + Specialist/SIU separation.

Phase 9

Approval authority.

Phase 10

Azure AI Search RAG.

Phase 11

Human decision + audit.

Phase 12

Optional dossier.

Phase 13

Deployment and demo hardening.

121. MVP Priority if Time Is Constrained

Protect in this order:

Must Work

1. Document extraction
2. Claim understanding
3. Policy Schedule
4. Settlement rules
5. Estimate review
6. Claims routing
7. Specialist/SIU logic

Strong Differentiator

1. Policy/SOP RAG

Easy to Simplify

1. Missing-information generator
 2. PDF dossier
-

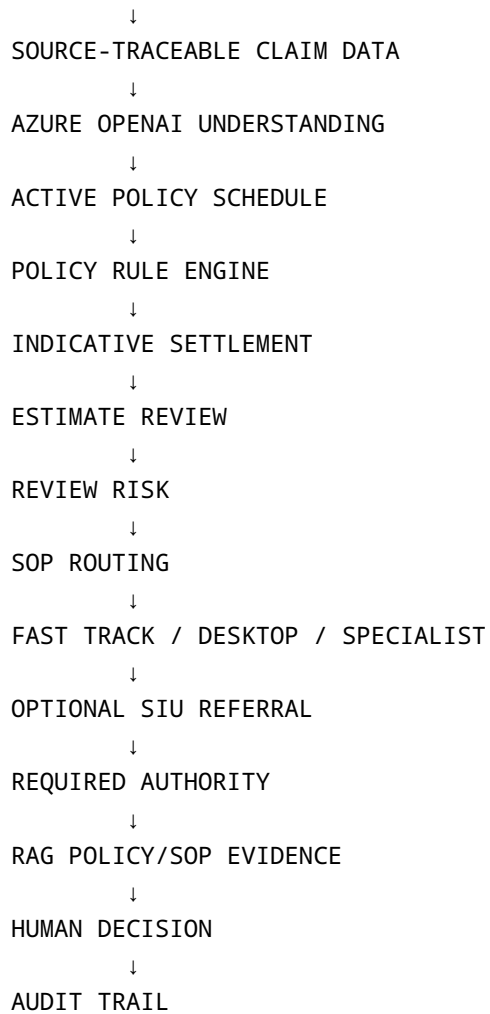
122. Definition of Done

ClaimIQ MVP is complete when this sequence works:

REAL CLAIM DOCUMENT

↓

AZURE DOCUMENT INTELLIGENCE



The application must visibly distinguish:

AI INTERPRETATION

POLICY RULE

SOP RULE

RAG EVIDENCE

SYNTHETIC BENCHMARK

HUMAN DECISION

No model-training claim may be made.

No anomaly may automatically be described as fraud.

No benchmark signal may automatically reduce settlement.

No RAG answer may invent a policy clause.

No AI recommendation may independently authorize, reject or repudiate a claim.

No SIU referral may be triggered by a risk score alone.

123. Final MVP Technology Stack

FRONTEND

Next.js
React
TypeScript
Tailwind CSS
shadcn/ui
PDF.js / React-PDF
Zod

DOCUMENT AI

Azure Document Intelligence
prebuilt-layout
REST API v4.0 / 2024-11-30

GENERATIVE AI

Azure OpenAI / Microsoft Foundry
OpenAI v1 Responses API
Structured Outputs

EMBEDDINGS

Azure OpenAI
text-embedding-3-small

RAG

Azure AI Search
Hybrid keyword + vector retrieval

BUSINESS DECISIONS

TypeScript deterministic rule engine

KNOWLEDGE

AD-COMP-01
AD-ELITE-02
CLM-MTR-01
CLM-MTR-02

DATABASE

None for MVP

HOSTING

Vercel

124. Product Story for the Board

ClaimIQ should not be presented as:

“We built an AI that predicts claims.”

Present it as:

“We redesigned claims assessment around AI-assisted exception management. Azure AI converts unstructured documents into usable claim data, insurer rules calculate what the policy allows, AI identifies what deserves attention, SOPs determine where the claim goes, RAG explains why, and humans retain final authority.”

That is the core product and transformation story.